

# Double Integrals



## Double integrals over a rectangle

- 1 Set up the double integral of  $f(x, y) = xy$  over the rectangle  $[0, 2] \times [0, 3]$  (integrate  $y$  first).
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## Iterated integrals

- 2 Evaluate  $\int_0^2 \int_0^3 xy \, dy \, dx$ .
- 3 Evaluate  $\int_0^1 \int_0^2 (x^2 + y) \, dy \, dx$ .
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## Fubini's Theorem: order of integration

- 4 Verify Fubini's Theorem for  $f(x, y) = xy$  on  $[0, 1] \times [0, 2]$  by evaluating the integral in both orders.
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## Double integrals over general regions

- 5 Evaluate  $\int_0^1 \int_0^x (x + y) \, dy \, dx$ .
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## Polar coordinates

- 6 Convert  $\iint_D (x^2 + y^2) \, dA$  to polar coordinates, where  $D$  is the disk  $x^2 + y^2 \leq 4$ , and evaluate.
- 7 Evaluate  $\iint_D xy \, dA$  where  $D$  is the quarter disk  $x^2 + y^2 \leq 1$ ,  $x \geq 0$ ,  $y \geq 0$ , using polar coordinates.
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## More iterated integral practice

- 8 Evaluate  $\int_0^1 \int_0^2 3x^2y \, dy \, dx$ .
- 9 Evaluate  $\int_0^1 \int_0^1 e^{x+y} \, dy \, dx$ .
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## The average value of a function

- 10 Find the average value of  $f(x, y) = xy$  over the rectangle  $[0, 2] \times [0, 4]$ , using  $\text{avg} = \frac{1}{\text{Area}} \iint_D f \, dA$ .
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## More general-region integrals

- 11 Evaluate  $\int_0^1 \int_0^{1-x} (x + y) \, dy \, dx$  (the triangle with vertices  $(0, 0)$ ,  $(1, 0)$ ,  $(0, 1)$ ).

- 12 Set up (don't evaluate)  $\iint_D x^2 y \, dA$  where  $D$  is the region between  $y = x^2$  and  $y = 2x$ , integrating  $y$  first (the curves meet at  $x = 0, 2$ ).
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### Finding area with a double integral

- 13 Use a double integral to find the area of the region bounded by  $y = x^2$  and  $y = 2x$ .
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### More polar-coordinate practice

- 14 Evaluate  $\iint_D \sqrt{x^2 + y^2} \, dA$  where  $D$  is the disk  $x^2 + y^2 \leq 9$ , using polar coordinates.
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### One more iterated integral

- 15 Evaluate  $\int_0^2 \int_1^3 (2x + y) \, dy \, dx$ .
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### A closing region-order problem

- 16 Sketch the region for  $\int_0^1 \int_{\sqrt{y}}^1 x \, dx \, dy$  and rewrite it with the order of integration reversed.
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### A final polar-coordinate application

- 17 Find the volume under  $z = x^2 + y^2$  and above the disk  $x^2 + y^2 \leq 1$ , using  $V = \iint_D (x^2 + y^2) \, dA$  in polar coordinates.
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### Setting up a volume integral over a rectangle

- 18 Set up (don't evaluate) the double integral for the volume under  $z = 4 - x^2 - y^2$  and above the rectangle  $[0, 1] \times [0, 1]$ .

- 19 Evaluate the volume integral set up above.
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### One last mass-density application

- 20 A flat plate occupying the rectangle  $[0, 2] \times [0, 1]$  has density  $\rho(x, y) = x + y$  (mass per unit area). Set up the double integral for its total mass.

- 21 Evaluate the total mass from the previous question.
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### One more general-region setup and evaluation

**22** Set up  $\iint_D y \, dA$  where  $D$  is the region bounded by  $y = 0$ ,  $y = x$ , and  $x = 4$ , integrating  $y$  first.

**23** Evaluate the integral set up above.

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### **A last polar-coordinate integral**

**24** Evaluate  $\iint_D 1 \, dA$  (the area) where  $D$  is the half-disk  $x^2 + y^2 \leq 4$ ,  $y \geq 0$ , using polar coordinates.